Q1.

Immigrant workers are M-specific, so the mobile labor is unchanged: $L = L_M + L_A$

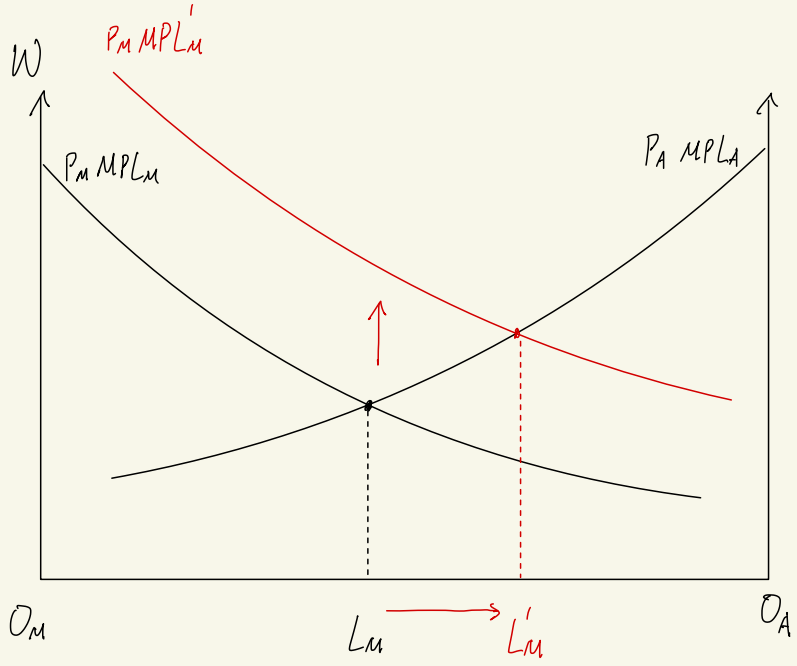
Competition among mobile labor implies: $W = P_M MPL_M = P_A MPL_A$

where $MPL_M = MPL_M(K, I, L_M)$ and $MPL_A = MPL_A(T, L_A)$

(+) (+) (−) — I: immigrants

(+) (−)

(a)



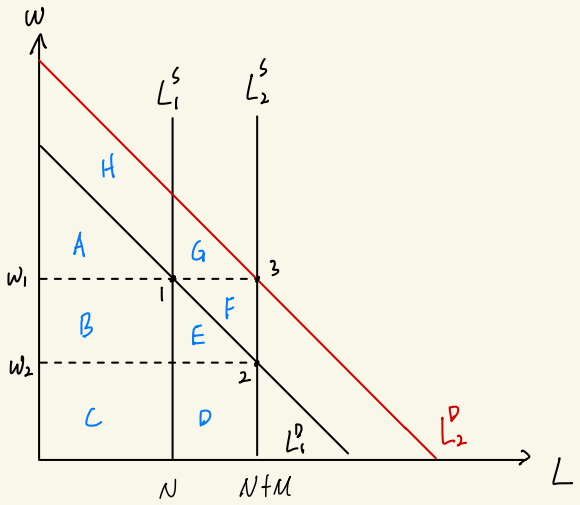
(b)

$W \uparrow$

$MPK \uparrow$ since $I \uparrow$, $L_M \uparrow$

$MPT \downarrow$ since $L_A \downarrow$

Q2



(a)

A : Initial capital income = $r_1 K_1$

B : Reduction in native workers' income due to immigration in the short run = $(w_1 - w_2) \times N$

C : Total income earned by native workers after immigration in the short run = $w_2 \times N$

D : Total income earned by immigrant workers in the short run = $w_2 M$

E : The immigration surplus : the increase in total output resulting from immigration that is earned by native-owned capital

F : The portion of immigrants' additional long-run income due to capital accumulation and increased labor demand.

G : The return to newly added capital in the long run.

H : The part of the return to old capital in the long run.

(b) Assumption : the long-run rate of return to capital returns to its initial level

$\Rightarrow$ capital can full adjust such that K/L remain unchanged.

(c).

Consider the CD production function : $Y = K^{\alpha} L^{1-\alpha}$

$w = MPL = (1-\alpha)\left(\frac{K}{L}\right)^{\alpha}$

$r = MPK = \alpha\left(\frac{K}{L}\right)^{\alpha-1}$

Immigrants enter : $L \uparrow \Rightarrow \frac{K}{L} \downarrow \Rightarrow w \downarrow , r \uparrow$

To restore $\frac{K}{L}$, K has to be adjusted : $K \uparrow$ until $\frac{K}{L}$ unchanged

If $\frac{K}{L}$ is unchanged, adverse wage effect disappears, and then the immigration surplus disappears.